

CN2025A-68

Galactic panel

Matthias R. Schreiber | matthias.schreiber@usm.cl

USM

faculty

Towards a complete and volume limited sample of Cataclysmic Variables: measuring orbital periods with SPECULOOS

Abstract

White dwarf binaries produce supernovae Ia and they might dominate the gravitational wave detections by space interferometers such as LISA. Despite recent progress, our understanding of the formation and evolution of these fascinating objects is rather limited. This is largely caused by the absence of a representative and large sample of white dwarf binaries that could be used to calibrate evolutionary models. However, the prospects of providing such a sample within the next few years are excellent. The most numerous white dwarf binaries are cataclysmic variables (CVs). All CVs up to a distance of 300-500pc (depending on the subtype) will be detected by eROSITA which offers the potential to establish a large volume limited sample. We run a large scale follow up program spectroscopically confirming CV candidates detected with eROSITA and subsequent characterization. Here we apply for measuring through photometric variability the orbital period of 22 CVs.

Observing Blocks

Instrument/Telescope	Req. time	Min. time	1 st Option	2 nd Option
CCD Specu- loos/Speculoos	22 nights	10 nights	April Any	July Any

Cols

Name	Institution	e-mail	Observer?
Jaco Brink	OnCL	jbrink@aip.de	True
Axel Schwope	OnCL	aschwope@aip.de	True
Sarai Zapata	USM	sarai.zapata@usm.cl	True

Status of the project

- Past nights: 19
- Future nights: 140

- Long term: True
- Large program: False
- Thesis: False

List of Targets

ID	RA	DEC	Mag
1eRASSJ122024.6-585041	12:20:24.34	-58:50:41.0	17.469086
1eRASSJ134035.3-430516	13:40:35.12	-43:05:14.8	17.062477
1eRASSJ140146.6-501321	14:01:46.69	-50:13:21.8	16.438608
1eRASSJ143410.6-500141	14:34:10.43	-50:01:40.7	18.46163
1eRASSJ144156.8-472618	14:41:56.88	-47:26:20.0	18.303326
1eRASSJ145600.4-531821	14:56:00.63	-53:18:18.9	19.436625
1eRASSJ151449.8-220059	15:14:49.57	-22:00:58.2	18.67135
1eRASSJ151504.8-523611	15:15:04.83	-52:36:13.2	17.243034
1eRASSJ152211.8-651507	15:22:11.28	-65:15:07.6	19.254879
1eRASSJ154026.0-233433	15:40:25.76	-23:34:34.9	18.641047
1eRASSJ154103.2-421035	15:41:03.50	-42:10:34.5	19.124432
1eRASSJ154608.6-324359	15:46:08.62	-32:43:55.9	18.452044
1eRASSJ161355.3-304640	16:13:55.43	-30:46:39.0	18.188858
1eRASSJ164623.7-194916	16:46:23.81	-19:49:17.1	19.022
1eRASSJ164641.8-421318	16:46:41.65	-42:13:21.0	19.281048
1eRASSJ165204.4-600002	16:52:04.49	-60:00:01.6	19.413431

SCIENTIFIC AIM AND RATIONALE

Cataclysmic variables (CVs) are the most common endproducts of close binary evolution (1). They consist of a white dwarf (WD) primary that accretes matter via Roche-lobe overflow (RLOF) from a low-mass companion (8). More than 90% of the systems have binary orbital periods below 4 hours and, in very general terms, they evolve under the influence of angular momentum loss (AML) through magnetic braking and gravitational radiation, from long to short orbital periods. Understanding close binary evolution is of imminent relevance to uncover pathways to SNIa and to understand the foreground signal of the planned space-based gravitational wave observatory. Yet there are fundamental unsolved problems concerning their sample properties and their evolution that shall be tackled by this proposal.

Accretion onto the WD is strongly dictated by its magnetic field. In CVs where the WD has a weak magnetic field an accretion disk forms that may extend to the surface of the WD, while in CVs with highly magnetized WDs (magnetic CVs, mCVs), known as polars and intermediate polars, there is at most a truncated disk and matter flows via accretion columns onto the magnetic poles of the WD. Observationally mCVs can be distinguished from non-magnetic systems by their emission lines profiles (symmetric disk lines for non-magnetic CVs, asymmetric lines from streams in mCVs), the strength of the He II $\lambda 4686$ emission line (similar to that of $H\beta$ in mCVs, weak or absent in non-magnetic CVs), and the occasional detection of cyclotron harmonic emission lines or Zeeman absorption lines as direct indicators of magnetism. Photometrically, most CVs show periodic variability on the binary orbital period (unless the orbital inclination is too small, which applies to a very small subset of the known population) and may show periodic variability also on the spin period of the accreting WD ($P_{\text{orb}} = P_{\text{spin}}$ for the polars, $P_{\text{spin}} \sim 0.1P_{\text{orb}}$ for the intermediate polars, flickering variability for the non-magnetic CVs.)

The evolutionary status of a given single system is written into its binary orbital period and the evolution of the class (and the sub-classes) is written into the binary orbital period distribution. Thus, the binary orbital period is *the* fundamental parameter to be determined for individual systems and complete samples to fully understand binary evolution.

There are two fundamental issues that were hampering progress in the past, one on the theoretical, one on the observational side. In the past, binary population synthesis (BPS) models (e.g. by 3) have completely neglected to take into account the effect of magnetic fields. It was shown by Belloni et al. (2), that magnetic fields will play an integral role in the evolution of compact binaries by weakening the effect of magnetic braking, and thereby slowing down the angular momentum loss (AML) in mCVs. Observationally only recently it has become clear that mCVs and non-magnetic systems have significantly different orbital period distributions (6), which implies that the strength of the magnetic field plays a significant role and forces mCVs to have different evolutionary pathways from non-magnetic systems. However, the strength of AML is not fully quantified so that a full-scale BPS that could be compared with observations is still missing.

On the observational side, sufficiently large complete samples are not yet available so that neither the space density nor the fraction of mCVs among all CVs are known. The space density has uncertainties of a factor of 2 or larger which is simply due to the small sizes of either flux- or volume-limited samples (7; 5). Pala's sample consists of 42 CVs, has an estimated completeness $>80\%$, and comprises a fraction of $\sim 36\%$ mCVs. However, a first eROSITA-selected flux-limited sample with 23 CVs contained only 2 magnetic systems (5).

The prospects to overcome the current observational limitations are excellent. All CVs are X-ray emitters (4) and this property allows to build large and complete volume-limited samples from the exploitation of the sensitive X-ray all-sky survey performed by eROSITA. From a probabilistic source match of eROSITA X-ray sources with Gaia objects a first batch of high-confidence CVs was selected. The target properties that led to their selection are described and illustrated in Fig. 2.

From the analysis of existing EFOSC2@NTT spectroscopy of 150 candidates the target association could be quantified to be better than $> 90\%$, as 133 new CVs were confirmed. eROSITA data allows to build a volume-limited sample of the various CV sub-classes out to 500 pc. The whole sample will comprise of about 1000 systems. Our subset of targets consist of likely mCVs for which time-resolved photometric observations are needed to sample the orbital period distribution of mCVs with better statistics than in (6). All the CVs in our target list are new to the literature and lack knowledge about their orbital periods. These can most easily be obtained from dedicated time-resolved photometric observations, that is proposed here. All the systems in our subset have equal priority for observations as no orbital period has been found for any of them.

We propose to perform time-resolved photometric observations of new confirmed CVs for which nothing more than their X-ray properties and an identification spectrum are known. Two example spectra obtained with EFOSC2 on NTT are shown in Fig. 1. The strong HeII emission line at $4686\text{\AA}$ clearly identify the object as a strong mCV candidates. The determination of the binary orbital periods of all the objects listed below is the one and only aim of this proposal. This measurement will deliver the missing piece of information to make use of the newly discovered CVs for sample studies and evolutionary considerations.

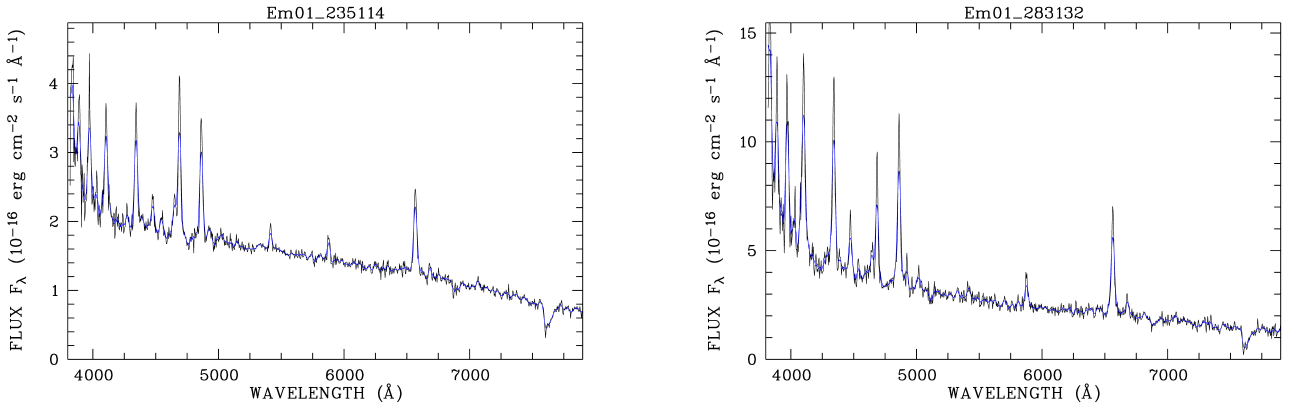


Figure 1: NTT identification spectra obtained with EFOSC2 of two of the proposed targets, showing very strong HeII $\lambda 4686$ emission lines, indicating the likely magnetic nature of the targets.

References

- [1] Belloni D., Schreiber M. R., 2023, in , Handbook of X-ray and Gamma-ray Astrophysics. Edited by Cosimo Bambi and Andrea Santangelo. p. 129, [doi:10.1007/978-981-16-4544-0_98-1](https://doi.org/10.1007/978-981-16-4544-0_98-1)
- [2] Belloni D., Schreiber M. R., Pala A. F., Gänsicke B. T., Zorotovic M., Rodrigues C. V., 2020, *MNRAS*, **491**, 5717
- [3] Goliasch J., Nelson L., 2015, *ApJ*, **809**, 80
- [4] Mukai K., 2017, *PASP*, **129**, 062001
- [5] Pala A. F., et al., 2020, *MNRAS*, **494**, 3799
- [6] Schreiber M. R., Belloni D., Schwöpe A. D., 2024, *A&A*, **682**, L7
- [7] Schwöpe A. D., 2018, *A&A*, **619**, A62
- [8] Warner B., 1995, Cataclysmic variable stars. Vol. 28

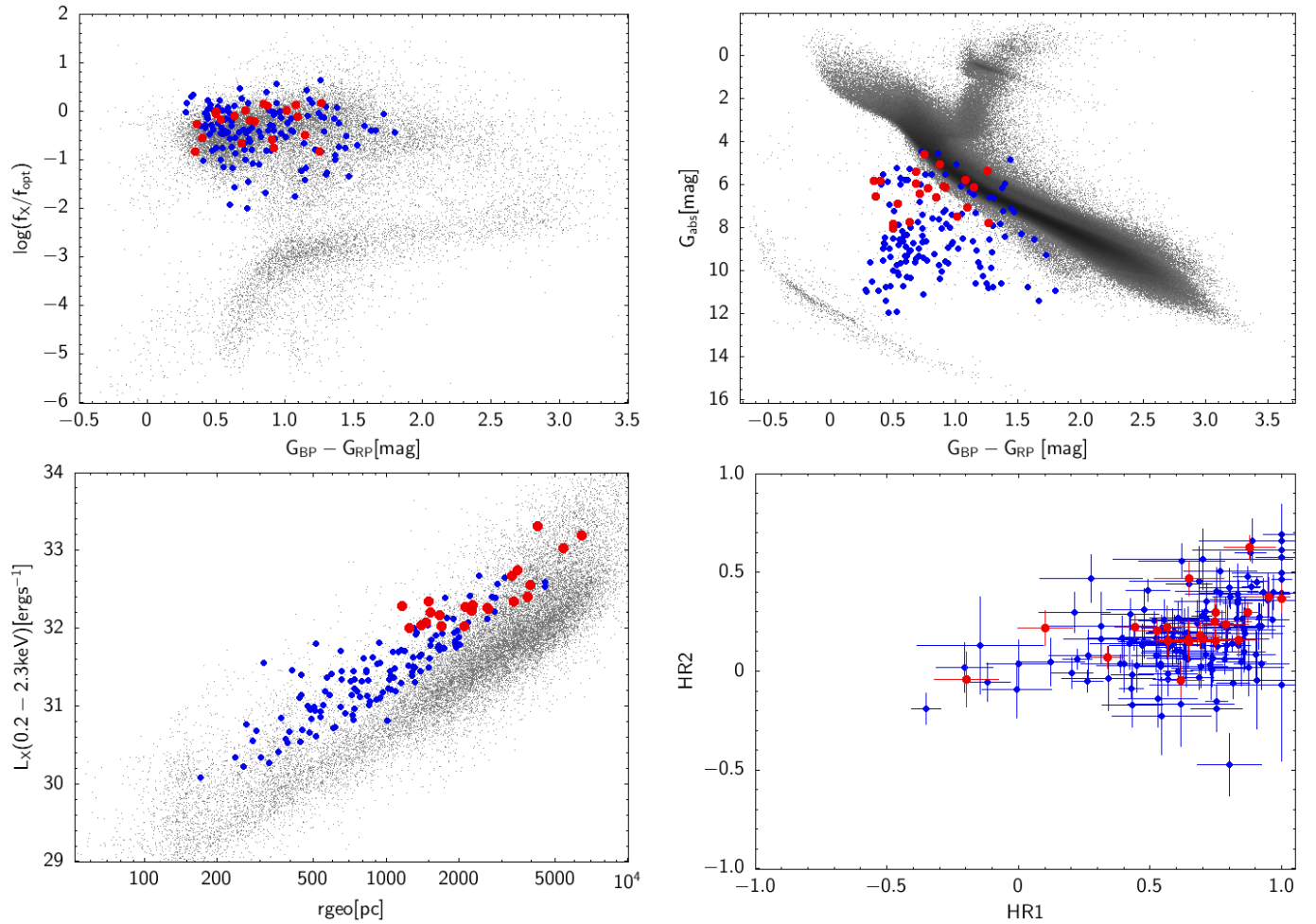


Figure 2: Diagnostic diagrams of newly identified CVs from eROSITA. Shown are an X-ray/optical color-color diagram, a color-magnitude diagram, a luminosity-distance diagram, and an X-ray color-color diagram. Shown in blue are all new CVs, those selected for this proposal are shown in red. The X-ray/optical color-color diagram shows the comparably high X-ray flux of the CVs (as high as those of AGN and 2-3 orders of magnitude higher than stars, both shown as grey background sources). The CMD illustrates the large spread in absolute magnitude thus covering all kinds of CVs (the grey background is made of non-X-ray emitting unrelated objects to outline the ZAMS and the WD sequence). The L_X -dist diagram illustrates the potential to build large complete samples out to 500 pc (grey background objects are selected from deeper eROSITA surveys under investigation in e.g. SDSS), and the X-ray hardness ratio diagram separates mCVs from non-magnetic ones through their soft X-ray spectra with $HR1 < 0.3$. A hardness ratio is defined as $HR = (H - S) / (H + S)$, with H and S being X-ray photons in two adjacent energy bands. With the given normalization, HR -values are constrained within $+1$ and -1 . An $HR1$ -value above 0.3 is compatible with just thermal emission from an accretion column or the disk-WD boundary layer. A value below 0.3 indicates the presence of a soft, blackbody-like source, as discovered from the heated photospheres in the vicinity of the accretion regions in mCVs.

TECHNICAL DESCRIPTION

For each of the targets in the list below we request time-resolved photometry during two successive nights with SPIRIT@SPECULOOS lasting at least 4 hours each. We choose the r filter for its high sensitivity. We plan to use integration times of 60 sec (for objects brighter than $G=17.5$ we will use 30 sec exposures). The observations shall be performed without interruption for the planned $\gtrsim 4$ hours per night. The first observation will give a first estimate of the binary orbital period. A second observation in the following night is requested to then determine the period with about 10% accuracy, sufficient for the later analysis in an evolutionary context. The second observation has to be placed in the following night after the first to not loose the cycle counts.

The target list consists of 22 targets and is compiled for observations beginning in April 2025, the observations shall last 6 month. For each target we request 2 observations, each lasting at least 4 hours. Given the duration of the nights in the observation period, we can therefore observe two targets per night, with each observation lasting at least 4 hours per target, and therefore request 22 nights of observations. While this would significantly slow down our progress, we could start the project with a smaller number of targets and slightly reduced observing time. However, allocating less than 10 nights (corresponding to 10 targets) would not make sense in the light of our ambitious goals. These targets form part of a larger list of targets for which time-resolved photometric observations are required to determine their orbital periods, and hence future SPECULOOS time will be requested as the project progresses. **Given the range of right ascensions covered by our target list, the requested 22 nights should be more or less homogeneously spread over the first portion of the observing period (April-July).**

Name	RA (deg)	DEc (deg)	G (mag)	distance (pc)
1eRASS_J122024m585041	185.10141884198	-58.84473049267	17.47	2641
1eRASS_J134035m430516	205.14631639061	-43.08745477089	17.06	1660
1eRASS_J140146m501321	210.44455008715	-50.22273264389	16.44	1159
1eRASS_J143410m500141	218.54345924208	-50.02795994939	18.46	1396
1eRASS_J144156m472618	220.48698370297	-47.43888285658	18.30	3836
1eRASS_J145600m531821	224.00262822874	-53.30525283517	19.44	2110
1eRASS_J151449m220059	228.70652408272	-22.01615394751	18.67	2276
1eRASS_J151504m523611	228.77010544185	-52.60367527565	17.24	1697
1eRASS_J152211m651507	230.54698822094	-65.25211164317	19.25	2254
1eRASS_J154026m233433	235.10733199197	-23.57634948551	18.64	6426
1eRASS_J154103m421035	235.26457765303	-42.17625569077	19.12	3932
1eRASS_J154608m324359	236.53592610533	-32.73218703467	18.45	3369
1eRASS_J161355m304640	243.48097358607	-30.777495679	18.19	4244
1eRASS_J164623m194916	251.59919641109	-19.82142928666	19.02	3330
1eRASS_J164641m421318	251.67353410031	-42.22249683314	19.28	3482
1eRASS_J165204m600002	253.01870286012	-60.00044240556	19.41	2133
1eRASS_J172757m324306	261.99056385458	-32.71885722903	19.15	2610
1eRASS_J174508m505151	266.28719889521	-50.86387745268	19.45	5416
1eRASS_J180106m454404	270.27628864167	-45.73491123791	16.62	1458
1eRASS_J180549m511557	271.45496339245	-51.26563582973	16.99	1530
1eRASS_J184412m424652	281.05288381169	-42.78106244822	17.40	1495
1eRASS_J190531m534943	286.38287025478	-53.82894611389	18.52	1252

JUSTIFICATION OF LONG-TERM STATUS OR WORK PLAN FOR LARGE PROGRAMS.

The current application is based on the analysis of a first batch of CVs from existing EFOSC2@NTT observations. These were selected on the basis of only the first of four complete X-ray all-sky surveys with eROSITA (Merloni et al, 2024, A&A 682, A34). The deeper data were stacked, counterparts were associated and a large-scale spectroscopic identification program was implemented in SDSS-V (Almeida et al., 2023, ApJS 267, 44). These SDSS observations are in full swing. Contrary to earlier phases of the SDSS survey, SDSS has gone all sky though modifications of the Las Campanas 2.5 m DuPont telescope. This is particularly relevant in the context of this proposal as it delivers southern targets. The southern CV sky is much less explored than the northern sky and many new CVs are to be discovered. All the new discoveries will have eROSITA X-ray and spectroscopic information but they are lacking detailed photometry that would allow the determination of their orbital periods. The Milky Way Mapper program of SDSS-V, that will run until 2027, is supposed to observe 1087 southern CV candidates brighter than $G = 20$ and within 500 pc, and mCVs at all distances. We estimate the completeness fraction to be 40% max. About half of the objects will be known or will have orbital period information from elsewhere, hence we will select of order 200 CVs that need time resolved photometry in the same manner as described here. This adds up to an equivalent of 160 nights to be distributed over the next 4 years. The result will be a cleanly characterized complete sample to constrain the evolutionary channels of accreting compact white dwarf binaries.

CURRENT STATUS OF THE PROJECT

In the last few years research performed by our group on magnetic white dwarfs and white dwarf binary evolution had a global impact. We showed that the evolution of magnetic white dwarfs in cataclysmic variables can be understood by incorporating reduced magnetic braking in the standard prescriptions for the evolution of these systems (1). We furthermore demonstrated that one of the scenarios frequently suggested for the generation of magnetic fields in white dwarfs, the common envelope dynamo model, fails to explain the observations of detached and semi-detached white dwarf binaries (2). We also published a comprehensive review on the X-ray emission of magnetic cataclysmic variables in ApJS (3) and a new idea for the magnetic field generation in white dwarfs in close binary stars in Nature Astronomy (4). Our partners are running a highly successful identification program of compact sources from the eROSITA surveys (5) which led to the discovery of the 2nd white dwarf pulsar (6) and, most importantly, allows to establish a large and volume limited sample of accreting white dwarf binaries. We have started a large-scale follow-up programs of these systems with the SDSS (7) and ESO telescopes. We here propose to measure the orbital period of systems discovered by eROSITA and confirmed through spectroscopy. The orbital period distribution of these systems is key to understand their evolution (8). Throughout the next four years we will combine eROSITA and SDSS discoveries of new CVs (led by Schwöpe and Schreiber) with follow-up observations (performed and analyzed by two PhD students, Jaco Brink from AIP and Sarai Zapata from UTFSM). This project plays a crucial role in the development of white dwarf binary research in the region of Valparaíso. In particular, combining detections of the European X-ray satellite (eROSITA) with follow-up observations using telescopes located in our country represents a fantastic win-win situation.

Relevant publications

- [P1] D. Belloni, M. R. Schreiber, A. F. Pala, B. T. Gänsicke, M. Zorotovic, and C. V. Rodrigues, “Evidence for reduced magnetic braking in polars from binary population models,” *MNRAS*, vol. 491, pp. 5717–5731, Feb. 2020.
- [P2] D. Belloni and M. R. Schreiber, “Are white dwarf magnetic fields in close binaries generated during common-envelope evolution?,” *MNRAS*, vol. 492, pp. 1523–1529, Feb. 2020.
- [P3] D. Belloni, C. V. Rodrigues, M. R. Schreiber, M. Castro, J. E. R. Costa, T. Hayashi, I. J. Lima, G. J. M. Luna, M. Martins, A. S. Oliveira, S. G. Parsons, K. M. G. Silva, P. E. Stecchini, T. J. Stuchi, and M. Zorotovic, “Breaking the Degeneracy in Magnetic Cataclysmic Variable X-Ray Spectral Modeling Using X-Ray Light Curves,” *ApJS*, vol. 256, p. 45, Oct. 2021.
- [P4] M. R. Schreiber, D. Belloni, B. T. Gänsicke, S. G. Parsons, and M. Zorotovic, “The origin and evolution of magnetic white dwarfs in close binary stars,” *Nature Astronomy*, Apr. 2021.
- [P5] A. Schwöpe, J. Kurpas, P. Baecke, K. Knauff, L. Stütz, D. Tubín-Arenas, A. Standke, S. F. Anderson, F. Bauer, W. N. Brandt, K. Covey, S. Demasi, T. Dwelly, S. Freund, S. Friedrich, B. T. Gänsicke, C. Maitra, A. Merloni, D. Muñoz-Giraldo, A. Rodriguez, M. Salvato, K. Stassun, B. Stelzer, A. Strong, and S. Morrison, “Compact white dwarf binaries in the combined SRG/eROSITA/SDSS eFEDS survey,” *A&A*, vol. 686, p. A110, June 2024.
- [P6] A. Schwöpe, T. R. Marsh, A. Standke, I. Pelisoli, S. Potter, D. Buckley, J. Munday, and V. Dhillon, “X-ray properties of the white dwarf pulsar eRASSU J191213.9–441044,” *A&A*, vol. 674, p. L9, June 2023.
- [P7] A. Schwöpe, K. Knauff, J. Kurpas, M. Salvato, B. Stelzer, L. Stütz, and D. Tubín-Arenas, “A first systematic characterization of cataclysmic variables in SRG/eROSITA surveys,” *arXiv e-prints*, p. arXiv:2407.20903, July 2024.
- [P8] M. R. Schreiber, D. Belloni, and A. D. Schwöpe, “The cataclysmic variable orbital period gap: More evident than ever,” *A&A*, vol. 682, p. L7, Feb. 2024.